```python
import pandas as pd
from sklearn.datasets import make_classification
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import precision_score, recall_score, roc_auc_score

# ------------------------
# CREATE IMBALANCED DATASET
# ------------------------

X, y = make_classification(
    n_samples=1000,
    n_features=10,
    n_classes=2,
    weights=[0.95, 0.05],
    random_state=42
)

df = pd.DataFrame(X)
df['Fraud'] = y

print("FIRST 5 RECORDS")
print(df.head())

print("\nCLASS DISTRIBUTION")
print(df['Fraud'].value_counts())

# ------------------------
# TRAIN TEST SPLIT
# ------------------------
```

```
PS C:\Users\Keerthi Reddy\Desktop\Fraud_Detection_Project> python project2.py
```

project2.py > ...

```python
# -------------------------
# TRAIN TEST SPLIT
# -------------------------

X = df.drop('Fraud', axis=1)
y = df['Fraud']

X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42
)

# -------------------------
# LOGISTIC REGRESSION
# -------------------------

lr = LogisticRegression(max_iter=1000)

lr.fit(X_train, y_train)

y_pred_lr = lr.predict(X_test)
y_prob_lr = lr.predict_proba(X_test)[:, 1]

print("\nLOGISTIC REGRESSION RESULTS")
print("Precision:", precision_score(y_test, y_pred_lr))
print("Recall:", recall_score(y_test, y_pred_lr))
print("ROC-AUC:", roc_auc_score(y_test, y_prob_lr))

#
```

```python
print("ROC-AUC:", roc_auc_score(y_test, y_prob_lr))

# -------------------------
# RANDOM FOREST
# -------------------------

rf = RandomForestClassifier(
    n_estimators=100,
    random_state=42
)

rf.fit(X_train, y_train)

y_pred_rf = rf.predict(X_test)
y_prob_rf = rf.predict_proba(X_test)[:, 1]

print("\nRANDOM FOREST RESULTS")
print("Precision:", precision_score(y_test, y_pred_rf))
print("Recall:", recall_score(y_test, y_pred_rf))
print("ROC-AUC:", roc_auc_score(y_test, y_prob_rf))

print("\nPROJECT COMPLETED SUCCESSFULLY")
```

```python
print("Precision:", precision_score(y_test, y_pred_rf))
print("Recall:", recall_score(y_test, y_pred_rf))
print("ROC-AUC:", roc_auc_score(y_test, y_prob_rf))

print("\nPROJECT COMPLETED SUCCESSFULLY")
```

PROBLEMS 6 OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\Keerthi Reddy\Desktop\Fraud_Detection_Project> pip install scikit-learn==1.0.2
scikit-learn==1.0.2) (1.3.2)
Requirement already satisfied: threadpoolctl>=2.0.0 in c:\users\keerthi reddy\appdata\local\programs\python\python37\lib\site-pa
s (from scikit-learn==1.0.2) (3.1.0)
PS C:\Users\Keerthi Reddy\Desktop\Fraud_Detection_Project> python project2.py
FIRST 5 RECORDS
```

| | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | Fraud |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 0 | 0.964799 | -0.066449 | 0.986768 | -0.358079 | 0.997266 | 1.181890 | -1.615679 | -1.210161 | -0.628077 | 1.227274 | 0 |
| 1 | 1.230350 | -0.566395 | 1.169487 | 0.831617 | -1.176962 | 1.820544 | -1.788927 | -0.984534 | -1.091341 | 0.209470 | 0 |
| 2 | 0.969347 | -0.432774 | 0.858089 | 0.793818 | -0.268646 | -1.836360 | -1.216083 | -0.246383 | -1.066667 | -0.297376 | 0 |
| 3 | 1.750412 | 2.023606 | 1.688159 | 0.006800 | -1.607661 | 0.184741 | -2.619427 | -0.357445 | -1.473127 | -0.190039 | 0 |
| 4 | -0.001184 | -0.711303 | 0.208114 | 0.117124 | 1.536061 | 0.597538 | -0.637329 | -0.939156 | 0.684698 | 0.236224 | 0 |

```
CLASS DISTRIBUTION
0    947
1     53
Name: Fraud, dtype: int64

LOGISTIC REGRESSION RESULTS
Precision: 0.6666666666666666
Recall: 0.16666666666666666
ROC-AUC: 0.8550531914893618

RANDOM FOREST RESULTS
Precision: 1.0
Recall: 0.5
ROC-AUC: 0.8339982269503547

PROJECT COMPLETED SUCCESSFULLY
PS C:\Users\Keerthi Reddy\Desktop\Fraud_Detection_Project>
```

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